

```
In [1]: import numpy as np
import pandas as pd
```

```
In [3]: data = {
    'Student_id': [1,2,3,4,5,6,7,8,9,10],
    'Name': ['Aryan', 'Anuj', 'Sufi', 'Samruddhi', 'Vaishu', 'Siddhi', 'Husain', 'Chinmay', 'Tanaya', 'Rosy'],
    'Age' : [20,20,np.nan,22,32,25,18,41,19,np.nan],
    'Gender' : ['Male', 'Male', 'Male', 'Female', 'Female', 'Female', 'Male', 'Male', 'Female', 'Female'],
    'Percentage' : [88,74,58,81,95,92,105,95,99,18],
    'Attendance' : [78,84,75,91,115,88,25,97,76,83],
    'Grade' : ['B', 'B', 'C', 'B', 'A', 'A', 'A', 'A', 'A', 'Z'],
}
```

```
In [5]: df=pd.DataFrame(data)
```

```
In [7]: df.head(10
)
```

```
Out[7]:
```

	Student_id	Name	Age	Gender	Percentage	Attendance	Grade
0	1	Aryan	20.0	Male	88	78	B
1	2	Anuj	20.0	Male	74	84	B
2	3	Sufi	NaN	Male	58	75	C
3	4	Samruddhi	22.0	Female	81	91	B
4	5	Vaishu	32.0	Female	95	115	A
5	6	Siddhi	25.0	Female	92	88	A
6	7	Husain	18.0	Male	105	25	A
7	8	Chinmay	41.0	Male	95	97	A
8	9	Tanaya	19.0	Female	99	76	A
9	10	Rosy	NaN	Female	18	83	Z

```
In [8]: df.isnull().sum()
```

```
Out[8]: Student_id    0
Name                0
Age                  2
Gender              0
Percentage          0
Attendance          0
Grade               0
dtype: int64
```

```
In [9]: df['Age'] = df['Age'].fillna(df['Age'].mean())
```

In [10]: df

Out[10]:

	Student_id	Name	Age	Gender	Percentage	Attendance	Grade
0	1	Aryan	20.000	Male	88	78	B
1	2	Anuj	20.000	Male	74	84	B
2	3	Sufi	24.625	Male	58	75	C
3	4	Samruddhi	22.000	Female	81	91	B
4	5	Vaishu	32.000	Female	95	115	A
5	6	Siddhi	25.000	Female	92	88	A
6	7	Husain	18.000	Male	105	25	A
7	8	Chinmay	41.000	Male	95	97	A
8	9	Tanaya	19.000	Female	99	76	A
9	10	Rosy	24.625	Female	18	83	Z

In [12]: df['Age']=df['Age'].astype(int)

In [13]: df

Out[13]:

	Student_id	Name	Age	Gender	Percentage	Attendance	Grade
0	1	Aryan	20	Male	88	78	B
1	2	Anuj	20	Male	74	84	B
2	3	Sufi	24	Male	58	75	C
3	4	Samruddhi	22	Female	81	91	B
4	5	Vaishu	32	Female	95	115	A
5	6	Siddhi	25	Female	92	88	A
6	7	Husain	18	Male	105	25	A
7	8	Chinmay	41	Male	95	97	A
8	9	Tanaya	19	Female	99	76	A
9	10	Rosy	24	Female	18	83	Z

In [15]: df[df['Percentage'] > 100]

Out[15]:

	Student_id	Name	Age	Gender	Percentage	Attendance	Grade
6	7	Husain	18	Male	105	25	A

```
In [16]: df[df['Attendance'] > 100]
```

```
Out[16]:
```

	Student_id	Name	Age	Gender	Percentage	Attendance	Grade
4	5	Vaishu	32	Female	95	115	A

```
In [20]: df[~df['Grade'].isin(['A','B','C','D','F'])]
```

```
Out[20]:
```

	Student_id	Name	Age	Gender	Percentage	Attendance	Grade
9	10	Rosy	24	Female	18	83	Z

```
In [21]: df['Percentage'] = df['Percentage'].apply(lambda x: 100 if x > 100 else (0 if x < 0
```

```
In [26]: df
```

```
Out[26]:
```

	Student_id	Name	Age	Gender	Percentage	Attendance	Grade
0	1	Aryan	20	Male	88	78	B
1	2	Anuj	20	Male	74	84	B
2	3	Sufi	24	Male	58	75	C
3	4	Samruddhi	22	Female	81	91	B
4	5	Vaishu	32	Female	95	100	A
5	6	Siddhi	25	Female	92	88	A
6	7	Husain	18	Male	100	25	A
7	8	Chinmay	41	Male	95	97	A
8	9	Tanaya	19	Female	99	76	A
9	10	Rosy	24	Female	18	83	F

```
In [23]: df['Attendance'] = df['Attendance'].apply(lambda x: 100 if x > 100 else (0 if x < 0
```

```
In [25]: df['Grade'] = df['Grade'].apply(lambda x : x if x in ['A','B','C','D','F'] else 'F'
```

```
In [29]: Q1 = df['Age'].quantile(0.25)
Q3 = df['Age'].quantile(0.75)

IQR = Q3 - Q1

lower_limit = Q1 - 1.5 * IQR
higher_limit = Q3 + 1.5 * IQR

df[df['Age'] > higher_limit]
```

Out[29]:

	Student_id	Name	Age	Gender	Percentage	Attendance	Grade
4	5	Vaishu	32	Female	95	100	A
7	8	Chinmay	41	Male	95	97	A

In [30]: `df['Age'] = df['Age'].apply(lambda x : df['Age'].median() if x > higher_limit else`

In [31]: `df`

Out[31]:

	Student_id	Name	Age	Gender	Percentage	Attendance	Grade
0	1	Aryan	20.0	Male	88	78	B
1	2	Anuj	20.0	Male	74	84	B
2	3	Sufi	24.0	Male	58	75	C
3	4	Samruddhi	22.0	Female	81	91	B
4	5	Vaishu	23.0	Female	95	100	A
5	6	Siddhi	25.0	Female	92	88	A
6	7	Husain	18.0	Male	100	25	A
7	8	Chinmay	23.0	Male	95	97	A
8	9	Tanaya	19.0	Female	99	76	A
9	10	Rosy	24.0	Female	18	83	F

In [32]: `minimum = df['Attendance'].min()
maximum = df['Attendance'].max()
df['Scaled Attendance'] = (df['Attendance'] - minimum)/(maximum - minimum)`

In [33]: `df`

Out[33]:

	Student_id	Name	Age	Gender	Percentage	Attendance	Grade	Scaled Attendance
0	1	Aryan	20.0	Male	88	78	B	0.706667
1	2	Anuj	20.0	Male	74	84	B	0.786667
2	3	Sufi	24.0	Male	58	75	C	0.666667
3	4	Samruddhi	22.0	Female	81	91	B	0.880000
4	5	Vaishu	23.0	Female	95	100	A	1.000000
5	6	Siddhi	25.0	Female	92	88	A	0.840000
6	7	Husain	18.0	Male	100	25	A	0.000000
7	8	Chinmay	23.0	Male	95	97	A	0.960000
8	9	Tanaya	19.0	Female	99	76	A	0.680000
9	10	Rosy	24.0	Female	18	83	F	0.773333

In []: